

Structure and Obstructions in the Rule 30 Eventual-Periodicity Problem

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Abstract

Wolfram’s Prize Problem 1 asks whether the center column of Rule 30’s lone-seed orbit is non-periodic. We leave Problems 1, 2 and 3 open. Instead, we map the provable/unprovable boundary. First, we fully characterize eventual period-one for all nonzero finite configurations: the two constant-trace fibers avoid finitely supported configurations off-origin, and the sharp constant-prefix horizon is $w+2$. Second, we delimit eventual period-two: a mechanized ω -automaton ladder — proved uniform in right depth, left depth, and diagonal bound — cannot close the case; an exhaustive forced-continuation census to $n = 24$ and $n = 30$ by a verified fast method shows exclusion hinges on a bounded set of extremal forcing states. Third, we catalog ninety-four failed attempts in nine obstruction classes, unified by one fact: Rule 30’s nonlinearity is its only difference from additive Rule 90, whose center column is eventually periodic, so every nonlinearity-free proof must fail. A route-by-route compendium and computational archive accompany the paper.

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1 Notation

Symbols recurring across Parts 3–5, defined again at first use in context.

F The Rule 30 map, (1).

$c_t = \text{Tr}(x)_t = F^t(x)_0$ The center column (trace) at time t .

R_j, L_j ($j \geq 0$) The right and left coordinates x_j, x_{-j} of a configuration.

$\mathcal{F}$ The finitely supported configurations.

$H_r(n), \gamma(n)$ Period-two census objects: the residue- r candidate sets and the extinction gap (Section 4).

RW, DLP, SEP, (PT2) Labels in the period-two reduction chain (6).

$\text{plain}(R), \text{pin}(R)$ Ladder language variants at right depth R (Section 5).

$k_\star$ The threshold past which the census departs from the spectral bound (Section 4).

2 Introduction

Rule 30 is the binary cellular automaton on $\{0, 1\}^{\mathbb{Z}}$ with local rule

$$F(x)_i = x_{i-1} \oplus (x_i \vee x_{i+1}), \quad (1)$$

left permutive: fixing x_i, x_{i+1} , the value $F(x)_i$ is a bijection of x_{i-1} .

Wolfram’s Rule 30 Prize poses three problems about the center column of the lone-seed orbit [11]: whether it is always non-periodic (Problem 1), whether each color occurs with equal asymptotic frequency (Problem 2), and whether computing its n th cell requires $\Omega(n)$ effort (Problem 3). The column is verified non-periodic only through 10^9 steps, and no finite check can settle Problem 1 — a limit we return to as obstruction H (Section 6). We address Problem 1 throughout, Problem 2 through the period-two thread, and Problem 3 only in the register.

The finite-configuration version of Problem 1 is open [6]: at most one column can be eventually periodic [4, Thm. 2b][3, Prop. 3], and the width-two column is never eventually periodic, though the technique does not extend to width one [5].

This paper is organized around one fact that we make load-bearing throughout. Rule 90, $F(x)_i = x_{i-1} \oplus x_{i+1}$, is also left permutive, yet its lone-seed center column *is* eventually periodic (identically zero off the initial step). Hence no argument that uses only left permutivity, additivity, or any property Rule 90 shares can decide Problem 1: it would prove a false statement about Rule 90. The single property Rule 30 has that Rule 90 lacks is the OR in (1), which latches. We call the requirement that an argument depend on this latch the *Rule 90 filter*. Every positive result below passes it; every failure catalogued in Part IV either violates it or hits one of eight further walls.

What is proved. Part I settles eventual period one (Section 3): both constant-trace fibers are computed in closed form, the finite configurations meet them only at the origin, and the horizon is sharp. Part III proves three limitation theorems for the period-two ladder (Section 5). These are honest negative results: they say a specific, natural, mechanized method cannot close period two, and why.

What is measured, and what is conjectured. Part II (Section 4) reduces period-two exclusion to a finite-word statement, carries the exhaustive census to $n = 24$ (and a verified fast method to $n = 30$), and reports a structural phenomenon: near extinction the surviving witnesses collapse onto $O(1)$ extremal forcing states sharing a long common suffix and a single failure symbol. We state this as a conjecture with its supporting data, never as a theorem.

What fails, and why. Part IV (Section 6) is the register. Nine obstruction classes account for the ninety-four routes attempted across this project. We give each class, the routes it kills, the kill-reason type, and its Rule 90 status. The exhaustive per-route list is the accompanying supplement.

Scope. Part V (Section 7) states precisely what is and is not reached, against the published state of the art, and lists the concrete open leads. We claim no progress on the prize problems themselves.

3 Part I: Finite configurations cannot generate a constant trace

This part is lifted, statement and proof, from a companion note carrying the full derivation, machine-checked Lean and SMT certificates, and a reproducibility appendix; only the two figures and the eight results are reproduced here.

We write $c_t := \text{Tr}(x)_t = F^t(x)_0$, $R_j := x_j$, $L_j := x_{-j}$ ($j \geq 0$), and let $\mathcal{F}$ denote the finitely supported configurations. The trace sees only column 0, but under a left-permutive rule a discrepancy left of the origin propagates rightward along a diagonal and must eventually reach it. The classification rests on this one fact, which uses only locality and left permutivity, not the specific form of (1).

Lemma 3.1 (Triangular uniqueness). *If two configurations agree at every coordinate $i \geq 0$ and have the same central trace, then they are equal.*

Proof. If $-n$ is their rightmost differing coordinate, locality preserves agreement to its right through time n . Left permutivity preserves the discrepancy along $(-n, 0), (-n+1, 1), \dots, (0, n)$, contradicting equal traces. In its finite form, if the rows agree on $i \geq 0$ and their traces agree for $0 \leq t \leq r$, the same induction forces agreement at $-r, -r+1, \dots, -1$. $\square$

Uniqueness alone does not say what the forced left half looks like. The next result computes it: the rows with zero trace form one invariant class $\mathcal{C}_m$ for each position m of the first right-hand one, each class is carried to its predecessor by F , and the left half is a prefix-OR of the right. Figure 1(a) is a member of $\mathcal{C}_3$.

Theorem 3.2 (Zero-trace fiber). *For a prescribed right half with $R_0 = 0$, the zero-trace-compatible left half is unique. If the positive half is nonzero and $m = \min\{j \geq 1 : R_j = 1\}$, the forced left half is $L_j = 0$ ($j < m$), $L_m = 1$, $L_j = j \bmod 2$ ($j > m$), with R_j for $j > m$ arbitrary. If instead the positive half is zero, the unique row is zero. Equivalently, in prefix-OR form,*

$$L_{2k+1} = \bigvee_{j=1}^{2k+1} R_j \quad (k \geq 0), \quad (2)$$

$$L_{2k} = R_{2k} \wedge \neg \bigvee_{j=1}^{2k-1} R_j \quad (k \geq 1). \quad (3)$$

Proof. For $m \geq 1$, let $\mathcal{C}_m$ consist of rows satisfying these conditions. One Rule 30 step gives

$$x'_0 = L_1 \oplus R_1, \quad R'_j = R_{j-1} \oplus (R_j \vee R_{j+1}), \quad L'_j = L_{j+1} \oplus (L_j \vee L_{j-1}). \quad (4)$$

Suppose $m > 1$. These identities give $x'_0 = 0$, $R'_j = L'_j = 0$ ($j < m-1$), $R'_{m-1} = L'_{m-1} = 1$, $L'_m = m \bmod 2$. For $j > m$, including $j = m+1$ because $L_m = 1$, the OR in the last update is one, so $L'_j = (1 - (j \bmod 2)) \oplus 1 = j \bmod 2$. Thus $F(\mathcal{C}_m) \subseteq \mathcal{C}_{m-1}$. For $m = 1$, the alternating left half is fixed, $x'_0 = 1 \oplus 1 = 0$, and $R'_1 = 0 \oplus (1 \vee R_2) = 1$ independently of the right tail; hence $F(\mathcal{C}_1) \subseteq \mathcal{C}_1$. Every $\mathcal{C}_m$ therefore realizes the zero trace. A nonzero right half selects a realization by its first one, and Lemma 3.1 makes it unique; a zero right half forces a zero left half by the same lemma against the zero row. $\square$

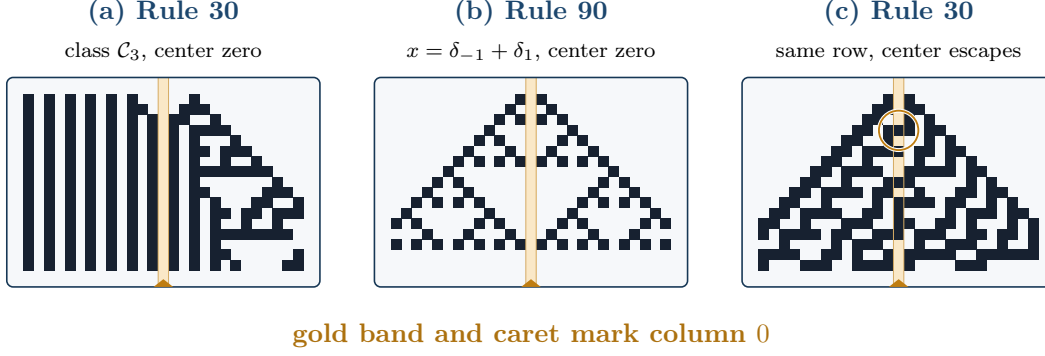


Figure 1: Time runs downward. In (a) the forced left tail alternates, hence is infinite, which is what Corollary 3.4 exploits. Panels (b) and (c) run one finite row under both rules; the difference is the OR latch in (1).

The other constant trace has a simpler fiber: one row per right half, and the left half no longer depends on which right half it is.

Theorem 3.3 (All-one fiber). *For a prescribed right half with $R_0 = 1$, the left half compatible with an identically-one trace is unique and independent of the right half: $L_j = 1 \iff j$ is positive and even.*

Proof. Write $L_0 := x_0 = R_0 = 1$, so the third identity of (4), $L'_j = L_{j+1} \oplus (L_j \vee L_{j-1})$, applies at every $j \geq 1$. Under the checkerboard pattern it gives $0 \oplus (1 \vee 0) = 1$ for even j and $1 \oplus (0 \vee 1) = 0$ for odd j , so the left half is fixed by F whatever the right half is. At the origin, $F(x)_0 = L_1 \oplus (L_0 \vee R_1) = 0 \oplus 1 = 1$ independently of R_1 , so the trace is identically one and $L_0 = 1$ is restored at every step. Lemma 3.1 makes the left half unique. $\square$

Every nonzero row in Theorem 3.2 carries ones arbitrarily far to the left, so none of them is finitely supported. That is the whole of the next statement, once one checks that a finite orbit cannot become the zero row.

Corollary 3.4. $\text{Tr}^{-1}(0^{\mathbb{N}}) \cap \mathcal{F} = \{0\}$. *Moreover, no column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually zero.*

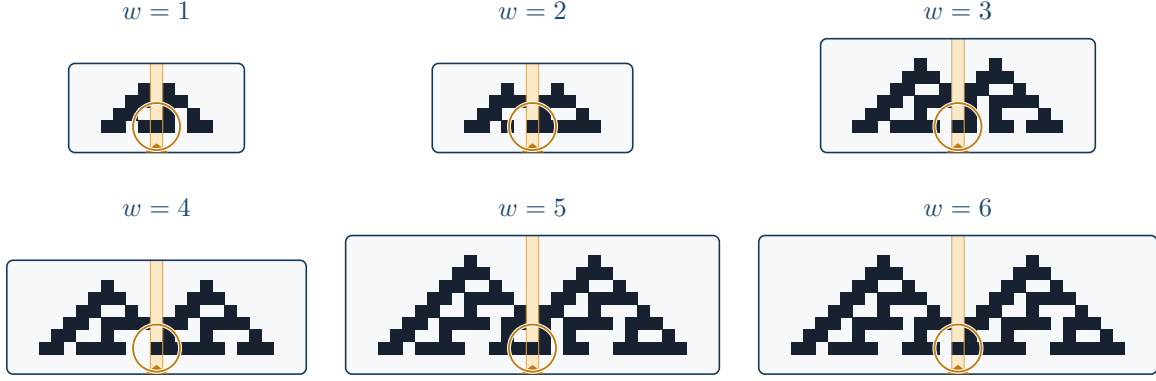
Proof. Every nonzero row in Theorem 3.2 has ones at all sufficiently large odd left depths. If a column of a finite orbit were zero from time T onward, translation would make $F^T(x)$ a finite zero-trace row. It is nonzero: if $[a, b]$ is the support interval of a nonzero finite row, its next row has ones at $a - 1$ and $b + 1$ because Rule 30 maps 001 and 100 to one. $\square$

Corollary 3.4 is not a consequence of left permutivity alone: Rule 90 is also left permutive, yet the finite row with ones at ± 1 keeps a zero center forever (Figure 1(b,c)). The Rule-30-specific step is the OR latch $R'_1 = 1 \vee R_2 = 1$ in the proof of Theorem 3.2, which has no analogue in an additive rule.

The same latch upgrades the corollary from one constant to both, with no further machinery. Theorem 3.3 proves the second half again and independently, since the checkerboard pattern has ones at every positive even depth; the argument below is shorter and is the one to quote.

Corollary 3.5. *No column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually constant.*

Proof. The eventually-zero case is Corollary 3.4. Suppose instead some column is eventually one; translating, take it to be column 0, so $c_t = 1$ for all $t \geq T$. Applying (1) at the origin gives $l_t = c_{t+1} \oplus (c_t \vee r_t)$, and $c_t = 1$ collapses the disjunction, so $l_t = 1 \oplus c_{t+1}$. For $t \geq T$ we also have $c_{t+1} = 1$, hence $l_t = 0$. Column -1 is then zero from time T onward, which Corollary 3.4 forbids. $\square$



circled cell = first nonzero center, at time $2\lceil w/2 \rceil + 1$

Figure 2: One extremizer of Theorem 3.6 per radius, truncated from the zero-trace completion at left depth w . The horizon is flat across each pair $w = 2k - 1, 2k$, which is the parity effect in the statement.

Corollary 3.4 says a finite row's center cannot stay zero forever. How long it can stay zero, as a function of the support radius, is exactly one step past the light cone of the first odd depth beyond that radius. For $w \geq 1$ let $\mathcal{S}_w = \{x \neq 0 : \text{supp}(x) \subseteq [-w, w], x_0 = 0\}$, and for $x \in \mathcal{S}_w$ let $H(x) = \max\{h : F^t(x)_0 = 0 \text{ for } 0 \leq t \leq h\}$.

Theorem 3.6 (Sharp finite horizon). *Let $w \geq 1$. Then*

$$\max_{x \in \mathcal{S}_w} H(x) = 2 \left\lceil \frac{w}{2} \right\rceil, \quad \#\left\{x \in \mathcal{S}_w : H(x) = 2 \left\lceil \frac{w}{2} \right\rceil\right\} = 2^w - 1. \quad (5)$$

Proof. Let q be the least odd integer greater than w , so $q - 1 = 2\lceil w/2 \rceil$. If the positive right half is nonzero, a trace zero through time q , compared with its zero-trace completion, would force by finite-prefix uniqueness and the prefix-OR identity $x_{-q} = \bigvee_{j=1}^q R_j = 1$, contradicting $q > w$. If that half is zero, comparison with the zero row makes every nonzero left cell visible by time w . Thus $H(x) \leq q - 1$. Conversely, choose any of the $2^w - 1$ nonzero words $(R_1, \dots, R_w)$, install the prefix-OR pattern through left depth w , and zero all other cells. This row agrees with its infinite zero-trace completion through the light cone of time $q - 1$; when w is odd, the extra depth $w + 1$ is even and forced to zero. The missing odd cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers. $\square$

The one-trace horizon is the same computation against the checkerboard fiber, and the parities exchange roles: ones sit at even depths, so the binding depth is the least *even* one past the radius, and $R_0 = 1$ already makes the row nonzero, which is why no extremizer is lost here.

Theorem 3.7 (Sharp finite horizon, one). *Let $w \geq 1$, let $\mathcal{T}_w = \{x : \text{supp}(x) \subseteq [-w, w], x_0 = 1\}$, and for $x \in \mathcal{T}_w$ let $H_1(x) = \max\{h : F^t(x)_0 = 1 \text{ for } 0 \leq t \leq h\}$. Let q be the least even integer greater than w . Then $\max_{x \in \mathcal{T}_w} H_1(x) = q - 1$ and $\#\{x \in \mathcal{T}_w : H_1(x) = q - 1\} = 2^w$.*

Proof. A trace one through time q , compared with the all-one completion of Theorem 3.3, would force by finite-prefix uniqueness $x_{-q} = 1$, contradicting $q > w$; thus $H_1(x) \leq q - 1$. Conversely, choose any of the 2^w words $(R_1, \dots, R_w)$, install the checkerboard pattern through left depth w , and zero all other cells. This row agrees with its all-one completion through the light cone of time $q - 1$; when w is even, the extra depth $w + 1$ is odd and forced to zero. The missing even cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers. $\square$

Combining the two horizons, the sharp law over both constants is flat in the parity that splits each one separately.

Corollary 3.8. *Let $w \geq 1$ and let $x \neq 0$ have $\text{supp}(x) \subseteq [-w, w]$. The longest constant prefix of the central trace of x has length at most $w + 2$, and exactly 2^w configurations attain length $w + 2$ when w is even, while exactly $2^w - 1$ attain it when w is odd.*

Proof. For $x_0 = 0$ Theorem 3.6 gives $2\lceil w/2 \rceil$, which is w for even w and $w + 1$ for odd w ; for $x_0 = 1$ Theorem 3.7 gives $q - 1$, which is $w + 1$ for even w and w for odd w . Hence the maximum horizon index is $w + 1$, so the maximum prefix length is $w + 2$. It is attained in the one case when w is even and in the zero case when w is odd, with the corresponding counts 2^w and $2^w - 1$. $\square$

The Rule-30-specific step is the OR latch in the proof of Theorem 3.2: the same finite row under Rule 90 keeps a zero center forever, so Corollary 3.5 is not a consequence of left permutivity. This is the Rule 90 filter passing on a proven result, and the template for reading every later obstruction.

4 Part II: The period-two problem — reduction, census, and a lead

4.1 The reduction chain

By Corollary 3.5 the constant traces are already excluded, so the only remaining candidate counterexample is the alternating phase $0101\dots$; period-two exclusion reduces, through a chain of proved uniform implications, to a single open separator. Writing (PT2) for “no nonzero finite configuration has $\text{Tr}_0(y) = \text{Tr}_0(F^2y)$,” the chain is

$$\gamma(n) \geq 1 \ \forall n \Rightarrow H_r(n) = \emptyset \Rightarrow \text{RW} \Leftrightarrow \text{DLP} \Rightarrow \text{SEP} \Rightarrow (\text{PT2}), \quad (6)$$

in which every arrow is proved and only the terminal separator SEP is open. Consequently a proof of $\gamma(n) \geq 1$ for all n would yield the first per-period exclusion theorem for the Rule 30 center column; the finite range is already discharged (below).

4.2 The forced-continuation census

A source word $W \in \{1, 2\}^n$ is extended by a deterministic forcing to a continuation that must remain hard-core; $\text{max-surv}(n)$ is the most forced rows any W survives and $\gamma(n) = (n + 2) - \text{max-surv}(n)$. The exhaustive range is pushed past brute force’s practical $n \sim 20$ – 22 ceiling by a dependency-edge dedup method: two source words reaching the same forcing state are provably indistinguishable to every later step, so the 2^n enumeration collapses to a walk over distinct states. The method is verified bit-for-bit against the original recursion and cross-checked exactly against brute force at every $n \leq 24$ with zero mismatches, a $\sim 17\times$ speedup already at $n = 18$.

Proposition 4.1 (Census, exhaustive and verified). *$|H_r(n)| = 0$ exhaustively for $n = 1, \dots, 24$, all residues and both constants; the rotated-wedge SAT encoding is UNSAT to $n = 30$; and $\gamma(n)$, computed by a fast dedup method cross-checked against brute force through $n = 24$ with zero mismatches, satisfies $\gamma(n) \in [7, 14]$ for $17 \leq n \leq 30$ with no monotone trend.*

Remark 4.2. The apparent “climbing” trend in the even- n subsequence is an artifact of sampling only even n ; the interleaved series shows no trend. One outlying cell ($n = 28$, $c = 2$) is bracketed by its neighbors as a statistical spike, not a signal, and is excluded below.

4.3 A structural lead: extinction is a needle, not a fog

Observation 4.3 (Terminal-class collapse; exhaustive $n \leq 18$). The words attaining $\text{max-surv}(n)$ number between two and eighteen, fall into one to four continuation classes, and within a class share a

common suffix of length $n - O(1)$, produce the identical forced continuation, and fail on the identical symbol. The forcing's live state is identical across a class (or differs only in the newest one or two cells). The phenomenon persists under restriction to hard-core sources, where the extremal words are exactly the clean periodic adversaries.

The extremal continuations are concrete and recognizable: hard-core words heavy in 2s and nearly periodic, e.g. 22222122222 and $2(221)^4$ — the forcing sits in a periodic orbit of the local map until a single phase slip lands. Two supporting facts are proved: a suffix lemma (words agreeing on a suffix from index m and on the first $2m$ forced symbols share their continuation forever; verified with zero exceptions) and that the sole Rule-30-specific ingredient of the forcing is the collapse $\Phi(2, \cdot) = \Phi(3, \cdot)$ of two left-actions, which is vacuous for Rule 90.

Conjecture 4.4. For all n , $\gamma(n) \geq 1$; equivalently, no reachable forcing state at source length n survives $n + 2$ forced rows. The extremal states, $O(1)$ per length, are the whole content.

The last clause of Observation 4.3 — that under restriction to hard-core sources the extremal words are the clean periodic adversaries — was the paper's one untried comparison, and it has since been tested and does not survive. Enumerating every word attaining $\max\text{-surv}(n)$ for $8 \leq n \leq 18$, with the per-tail maxima agreeing exactly with the project's registered table, the genuinely short source periods $(1, 3, 4, 5, 6, 7)$ occur only through $n = 11$; from $n = 12$ the smallest period present in the extremal set climbs with n , and a period of $n - 1$ or $n - 2$ on a length- n word is not periodicity but the tautology that a suffix matches a prefix. The single-defect shape $2^a 12^b$ accounts for four of the eleven extremal words at $n = 8$, one at $n = 9$, one at $n = 11$, and *none* for $n \geq 12$. The counting claim of the observation is unaffected: the number attaining the maximum stays inside $[2, 18]$ throughout.

The consequence is that the extremal sources are not a bounded-period family, so extinction on a finite list of periodic rays — the shape that would have carried a finite automaton or a one-dimensional recurrence — is not available. Extremal *continuations* behave better, keeping small periods with only occasional aperiodic cases, but that is a materially weaker statement and is not what the finite-ray argument required. Conjecture 4.4 itself is untouched; what is removed is the route to it that Observation 4.3 appeared to offer.

4.4 Why first-moment and spectral bounds cannot close it

Proposition 4.5. *The hard-core transfer matrix has spectral radius exactly $\varphi = 1.618\dots$, so the induced bound $\lambda < 2$ holds vacuously at every width and cannot witness extinction (which needs $\lambda < 1$). The bound is tight for $k < k_\star$ and off by a factor ≈ 300 at extinction, so no finite-width transfer or first-moment argument closes the census; the departure past $k_\star$ is the whole phenomenon.*

The natural first-moment repair also fails, for a structural reason rather than a computational one. The measured per-row survival rate $\varphi/4 \approx 0.4045$ is exact, not fitted: φ^k counts the hard-core (Fibonacci) language a surviving continuation must land in, and 4^k is the codomain of the forced-symbol map. If that map were equidistributed, fiber-counting alone would force $\gamma(n) \rightarrow \infty$ with no pseudorandomness assumption; it is not equidistributed, since the map's image saturates at a fixed fraction of 2^n well below 4^k and its fibers stabilize at a small constant rather than shrinking, so past $k \sim n/2$ the argument would need a fiber smaller than one, which is impossible for a nonempty fiber. Both bounds fail for the same underlying reason: the forcing recursion has genuinely unbounded memory (every new symbol rewrites the entire stored state), and no bounded-memory model, spectral or first-moment, can represent an unbounded one at any finite width.

5 Part III: The periodicity ladder and its limits

The ladder encodes “the center column is eventually p -periodic” as ω -automaton emptiness over column pairs, with the inverse transduction, a light-cone wedge, and a Buchi periodicity condition. An EMPTY verdict at some finite depth would deliver $\text{Thm}(p)$ directly: together with the classical fact that two adjacent columns cannot both be eventually periodic [3, Prop. 3][5], no admissible lasso would mean no lone-seed diagram has an eventually p -periodic center column, for any transient length. Calibration mechanically reproduces Part I’s period-one theorems at the depths where they apply, and separately passes the Rule 90 soundness control, which stays NONEMPTY throughout since Rule 90’s center column is genuinely eventually periodic. For $p = 2$ the language is nonempty at every decided depth, and we prove this is not a computational limit but a structural one: three independent levers of the encoding are each shown incapable of forcing emptiness, by proof rather than by exhausting compute.

Theorem 5.1 (Three inert levers). *For the $p = 2$ ladder: (i) the accepting verdict is uniform in the right depth R over a range of raw sizes exceeding $5000\times$; (ii) the boundary pin is sandwiched, $\text{plain}(R + 1) \subseteq \text{pin}(R) \subseteq \text{plain}(R)$, so it decides nothing depth does not; and (iii) the accepting tail system’s bisimulation-class count is exactly constant in the left depth k (15, 20, 32 in the three encodings, over a $3068\times$ range of raw states), because every accepting state lies in the saturated region where the left-depth constraints are inert. No lever of the ladder can force emptiness.*

Each lever closes by an exact argument, not only by measurement. The sandwich $\text{plain}(R + 1) \subseteq \text{pin}(R) \subseteq \text{plain}(R)$ holds because the boundary pin is exactly the solvability condition for the one column the strip does not model, and is therefore entailed by any genuine extension one column further out: it can only find, one column early, an emptiness that depth would find anyway. The left-depth invariance holds because the ladder’s own saturation rule stops checking the light-cone wedge once the letter count passes a fixed threshold, so every column added beyond that threshold affects neither of the two values the automaton returns; the accepting tail therefore lives entirely in a region where left depth is provably inert, not merely observed to be flat. Both closures apply equally to the Rule 90 control, which is what keeps the soundness check alive: Rule 90’s own extendability condition is vacuous, so its ladder is the plain language and never inherits Rule 30’s pin.

Proposition 5.2 (The witness-extension test is invalid). *Extracting a ladder witness and testing whether it extends to a genuine lone-seed diagram cannot decide anything: on the Rule 90 control, where a genuine witness provably exists, the sampler misses it by a factor of four in depth. The nonemptiness is attributable to the free boundary of the truncated strip (obstruction F).*

The Rule 90 control is what condemns the test, not a property of the Rule 30 sample. A genuine, fully realizable Rule 90 witness exists at onset one and extends to the probe limit, yet randomized sampling of the accepting component never finds anything within a factor of four of it, at every tested depth. Since the sampler already fails on an object known to be genuine, its failure on the Rule 30 candidates carries no information about whether those candidates are spurious.

6 Part IV: The register of obstructions

Ninety-four distinct routes have been attempted against Problems 1–3 in this project.¹ Nine obstruction classes account for essentially all of them. We state each as the reason a route dies, name the kill-reason type, and give its Rule 90 status. The exhaustive per-route register is the supplement.

¹Seventy-six from the primary tracking log, eighteen merged from a second, independently numbered log. One row number, 76, was assigned to an unrelated route in each (S-adic subword complexity in the first; the d -uniform survivor invariant in the second); both are retained under that number and distinguished by stream in the supplement.

- A. The $O(\log t)$ propagation wall.** Trace-anchored analysis determines a region of depth only $O(\log t)$, from one-bit-per-step information loss; three independent representations hit it. *Type:* depth. Does not bind the period-two counting question, which uses the periodicity hypothesis globally. The rate is a statement about the *right* light cone, where the local rule reduces to a running parity and the diagonal periods double. The left cone reaches far deeper for the same period budget and A should not be quoted there with this rate; see Remark 6.2.
- B. The Rule 90 filter.** Any argument using only left permutivity, additivity, or expansivity proves a false statement about Rule 90. *Type:* rule-generic. The organizing obstruction of the whole paper.
- C. Single-column blindness.** Entropy- and complexity-boundary framings measure a two-dimensional field insensitive to any one column. *Type:* population/observable. Kills the Kolmogorov and block-entropy routes.
- D. The missing composition law.** Every proposed Problem 3 shortcut needs a composition law it never supplies. *Type:* composition.
- E. The measure-zero single-orbit gap.** Rule 30 is ergodic for Bernoulli measure, so almost-everywhere density results never see the lone seed. *Type:* measure. Kills the density routes to Problem 2.
- F. The free boundary of a fixed-depth strip.** Any fixed-depth column window leaves its outermost column unconstrained, and the phase slip relocates there. *Type:* boundary. The ladder's wall (Part III).
- G. Arbitrary-input measures versus a fixed point.** Circuit size, decision depth, sensitivity and the like are identically zero on one fixed input. *Type:* object-mismatch. Kills the Problem 3 complexity routes.
- H. Finite certificates are complexity lower bounds, never more.** Any bounded exclusion or census bounds a complexity function and cannot become an all-length theorem. *Type:* intrinsic. Why no census closes a theorem.
- I. The pinned light-cone CNF.** With n fixed, the circuit CNF refutes only that instance. *Type:* object-mismatch. Kills the pinned Problem 3 SAT.

Remark 6.1 (A named gap: non-invertibility). Two routes (rows 60, 62) die by a mechanism none of A–I names: Rule 30 has no inverse, so it admits no unitary evolution operator and its defect world lines merge and annihilate rather than braid. This is a real, distinct kill reason, not a restatement of the Rule 90 filter (B) — it is a structural fact about Rule 30 itself, not an argument that also proves something false about Rule 90. We record it here rather than retrofit a tenth letter, since no other row in the register was found to die by it.

Remark 6.2 (The ordered wedge, and why obstruction A has two faces). Write the local rule in each light-cone frame. With $R_j[t] = s(t, t - j)$ one gets $R_j[t] = R_j[t - 1] \oplus (R_{j-1}[t - 1] \vee R_{j-2}[t - 1])$, in which R_j is the running parity of a function of strictly smaller indices; this is the induction behind the power-of-two right-diagonal periods, and behind obstruction A. With $L_j[t] = s(t, -t + j)$ one gets instead

$$L_j[t] = L_{j-2}[t - 1] \oplus (L_{j-1}[t - 1] \vee L_j[t - 1]),$$

where the OR is self-referential and absorbing: when $L_j[t - 1] = 1$ the update collapses to $L_j[t] = \neg L_{j-2}[t - 1]$ and the middle argument is discarded. The two frames behave oppositely, and the left one is the tame side.

That the left region is ordered is not new: [8, p. 871] records that L_j is eventually periodic with the first occurrence of period 2, 4, 8, 16, 32, 64 at depth 3, 8, 29, 400, 87 867 and 2 107 985 255 or more. We reproduce that table independently, to the unit, and it is what pins the identification of L_j with Wolfram’s diagonal j cells in from the left edge. What the frame adds is the recurrence above, which makes the asymmetry mechanical rather than pictorial, and the observation that left-frame periodicity of period g to depth D is exactly the spacetime isometry

$$s(t + g, x - g) = s(t, x) \quad \text{on } -t \leq x \leq -t + D,$$

verified directly in (t, x) coordinates on 5.5×10^6 cells with no exception, for g a multiple of 16 and for no other g tested. In that form the ordered region is the same kind of object as the periodic tori censused in Part III, which is the only structural bridge between them we know.

Two consequences for the register. First, obstruction A is a statement about the *right* cone. With period at most P the right frame reaches depth $\approx 2.4 \log_2 P$, while the left reaches 3, 8, 29, 400, 87 867, $\dots$ — super-exponential in $\log P$. A’s verdict survives, since both still demand unbounded period at depth $\Theta(t)$, but its rate is wrong on the left and should not be quoted there. Second, and this is why the remark belongs in a paper whose unifying fact is the Rule 90 filter: replacing $\vee$ by $\oplus$ destroys the phenomenon outright. The Rule 90 left frame is a running parity like its right one, its measured period is exactly j , and it has no ordered region and no order/disorder boundary. In this frame the OR is what creates order rather than what destroys it.

We record two cautions with it, both learned the hard way here. The measured region is ordered in *time* and aperiodic in *space* — its rows carry no spatial period up to 2048 on bands of width 2737 — so it is invisible in a plot. And a scan of it stopping at $j = 65\,536$ shows an unbroken period-16 plateau, which is not a law: the plateau ends at $j = 87\,867$. This is obstruction H in the form that is easiest to miss, since a plateau, unlike a certificate, looks like a theorem and costs nothing to state.

Remark 6.3 (The five sightings of the needle). The collapse of Observation 4.3 was recorded five times before, under five names, and dropped each time for a kill reason of type C or E — a population statistic — when the deciding object was never the population but the $O(1)$ extremal states. This is the methodological lesson of the register.

Table 1 is the compact register: all ninety-four routes, one line each. Rows marked $\dagger$ come from the second tracking log described above; row 76’s collision is kept under its original number in both. Full paragraphs, with exact quantities, are the accompanying supplement.

Table 1: The obstruction register: ninety-four routes.

#	Route	P	St.	Kill reason
1	R1: zero-set obligation (c eventually periodic $\Rightarrow r$ eventually periodic on $\{c_t = 0\}$)	1	Opn	Best route; kill condition not fired.
2	R2: cascade descent from the pin	1	Kil	Own kill condition fired: 1.14% of cells determined, density halves leftward, zero at $x \geq 1$.
3	R3: bounded-depth pin bootstrap (run-of-ones wedge)	1	Kil	Proved dead before remeasurement: bounded one-runs give only a bounded wedge, not a descent.
4	R4: right-boundary-anchored arguments	1	Kil	Right-boundary regularity reaches only $\sim 2.4 \log_2 t$ diagonals against a target at diagonal t .
5	R5: left permutivity / expansivity alone	1	Kil	Rule 90 filter: left permutive, lone-seed centre column eventually periodic (Kopra).
6	R6: complexity/entropy boundary framings	1,2,3	Kil	Literal version false in two lines; statistical rescue fails three ways; 2D entropy is column-blind.

#	Route	P	St.	Kill reason
7	R7: periodicity ladder, per-period exclusion	1	Opn	Depth, the boundary pin, and Q are all proved unable to close it (Cor. 2/3); blocked on one named ingredient, an aperiodic tail construction, which is obstruction F in its sharpest known form. Row 77 retired the other of the ladder's two modes outright, which is why only this one ingredient remains.
8	R8: invariant-measure classification of orbit closure	2	Opn	Consistent with unique ergodicity at every cell; not killed, not advanced. Conditional on row 79: if the checkerboard temporal fixed point lies in the orbit closure Y , this route's target fails outright; unbounded patch growth (row 79's open condition) is what keeps it alive.
9	R9: proof-complexity lower bound for c_n	3	Opn	Precondition probe ran (GMUS near-constant fraction); inverted outcome did not fire; bound unattempted.
10	Arm 1: Lean/evolutionary search for a P1 proof	1	NR	Declined: "find a lemma" is not pre-registerable.
11	Arm 2: search for the density-1/2 invariant measure	2	NR	Declined: pathway lands one measure-zero step short of the lone seed.
12	Arm 3: WASM tournament for sub-quadratic centre algorithm	3	Kil	Sealed LLM tournament never reached its decision band; a local follow-up extended the known-algorithm calibration to $n = 1,000,031$ instead, past the pre-registered threshold: exponent 1.98–2.03, no improvement found.
13	Arm 3 iter. 2: GF(2) seed grammar + Z3 rewrite	3	NR	Design only; peephole rewrites cannot connect $\Theta(n^2)$ composition to sub-quadratic.
14	Arm 4: AIR / frequency-domain shortcut	3	Kil	Pre-registered kill fired on first run: maximal linear complexity, exact ANF degree, plateauing density.
15	Arm 4 iter. 2: additive-FFT over $GF(2^k)$	3	NR	Design only; no root-of-unity structure in a binary field; no FiniteField sort in the SMT solver used.
16	Arm 5: Rule 150 kernel + sparse-error superposition	3	Kil	$N_{\text{eff}} \sim n^{1.68}$, decisively superlinear; escape hatch (row/block parity) also measured shut.
17	Arm 6: finite 2-kernel / bounded-rank digit query	3	Kil	Both pre-registered kills fired: ≥ 8191 residual sequences, GF(2) rank 512 at depth 9.
18	Arm 7: exact dyadic spacetime tile grammar	3	Kil	Diagnostic structure only; tile perimeters carry $\Theta(\text{scale})$ bits. Its own acceptance gate was already run through Crosstalk and closed as row 19, same files, same criteria verbatim.
19	Exclusion-guided search: dyadic tile composition algebra	3	Kil	Zero surviving candidates across six critic rejections; synthesis over-reached.
20	Arm 8: ROBDD of the arbitrary-input transfer function	3	Kil	Exponential growth ($\sim 1.88\times/\text{step}$) in all five tested variable orders.
21	Exclusion-guided search: single-seed transfer monoid algebra	3	Kil	Truncated $GF(2)[x]/(x^k)$ candidate falsified: first mismatch at step 5.
22	Exclusion-guided search, run 1	3	Kil	Zero surviving candidates; four of five depend on a parity cancellation already measured false.
23	Exclusion-guided search, run 2, typed exclusions	3	Kil	Empty frontier; three rejections each needing an unproved or undefined lemma.
24	Exclusion-guided search, run 3: carry-polynomial transfer	3	Kil	Deterministic probe found scalar wrong at $n = 2$: candidate 1, ground truth 0.
25	Zero-trace fiber classification	1	Prv	No nonzero finite configuration has an identically-zero centre trace; sharp horizon $2\lceil w/2 \rceil$; false for Rule 90.
26	All-one trace fiber	1	Prv	All-one trace forces the checkerboard left half; with row 25, excludes every eventually-constant centre. The eventually-one special case was already Jen 1986 Thm. 7a (row 84); novelty here is componentwise, not this exclusion itself.

#	Route	P	St.	Kill reason
27	Phase-labelled generalization of the OR-latch	1	Kil	Falsified at radius one: centre-one phase removes rather than preserves the right-neighbor latch.
28	Inverse-trace continuation	1	Kil	Produces only per-case contradictions, never period-independent; a follow-on evolutionary search aimed at the missing uniform-tail step returned nothing (0/72 claims verified).
29	Periodic-mask contraction	1	Kil	Falsified for every period word containing a zero: discrepancy moves earlier, never erased.
30	Exclusion-guided search: period-uniform inverse-trace obstruction	1	Kil	Zero of 72 claims verified; synthesis empty; two Lean artifacts contained sorry .
31	Same-forward-orbit collision reduction	1	Opn	Equivalence proved; remaining obligation (no collision at any period) still needed, starting at $p = 2$.
32	Incremental SMT exhaustion of nonconstant periods	1	Opn	Exact horizon table plus three proved partial results; period-two reduces to a stroboscopic F^2 problem with two one-sided cases excluded, mixed support open — likely the same separator as the (PT2)/needle conjecture.
33	Right-cone bit-parallel reconstruction + support certificate	1	Kil	1824 sweeps show no descent; the prescribed follow-on (row 34) closes the line via obstruction A.
34	Right-cone diagonal periodicity	1	Kil	Periodic region reachable from the boundary is $\sim 2.4 \log_2 t$ diagonals wide; centre is at diagonal t .
35	Extending the constant-trace mechanism to nonconstant periods	1	Kil	Falsified by direct measurement: forced bit still differs at depth 512 for four period words.
36	Periodicity bridge: Kopra's width-two theorem to nonlinear class	1	Kil	Of eight quiescent left-permutive ECAs, the nonlinear ones with full support reduce to Rule 30 itself.
37	Periodicity bridge: finite-delay reconstruction	1	Kil	Explicit counterexamples at $p = 148, t = 1855$ and $p = 110, t = 13219$ with persistent adjacent-column defects.
38	Periodicity bridge: centre-trace injectivity	1	Kil	$\{0\}$ and $\{0, 1\}$ share a centre trace at every checked step; defect escapes rightward.
39	Periodicity bridge: doubling separation	1	Kil	First fails at $p = 4$, where both values are one.
40	Exclusion-guided search: period-independent contradiction	1	Kil	Zero candidates survived; ten rejects reduce to six recurring reasoning errors.
41	Alternating-trace fiber: bounded-depth exclusion certificates	1	Opn	Closed-form survivor automaton found, but certificate levels grow with d and cannot reach uniformity alone.
42	Overnight literature-novelty triage (ten techniques)	infra	Inf	Verdict table gating the overnight arms: 5 closed, 5 known-adjacent, 0 novel-open.
43	ANF/Walsh analysis of the iterated centre-bit function	—	Prv	$\deg f_t = 2t - 1$ for all $t \geq 3$; implies nothing for P1/P2, provably inert for P3.
44	Classification of jointly invariant Markov measures, memory ≤ 3	2	Stl	Decisive at $m = 1, 2$; Gröbner basis kill condition fired at $m = 3$ (110 min, no termination).
45	Anashin 2-adic ergodicity	2	Kil	Four fatal reasons, including: Rule 90 has identical structure, so the object does not separate the rules.
46	Ore-relation ladder toward non-2-automaticity	1	Opn	Reduced, not killed: each rung one-sidedly proved; stopped at one lemma needing a polynomial bound.
47	Single-seed density arm (martingale/counting/invariance)	2	Kil	Original mechanisms structurally invalid (no probability space on a deterministic orbit); two weak byproducts survive.
48	Persistent homology / TDA on the spacetime point cloud	1,2	Kil	Time-filtered H_1 bars biject with known white triangles; column periodicity constrains no neighborhood.
49	Quantum circuit mapping, entanglement entropy	1,3	Kil	Single-seed basis-state entropy across the centre cut is identically zero.
50	Hydrodynamic / coarse-grained PDE limit	2	Kil	Coarse-grained density is a spatial-average object; P2 is about one measure-zero orbit.

#	Route	P	St.	Kill reason
51	Non-standard analysis / hyperfinite rows	2	Kil	Conservative: needs the standard equidistribution lemma already, and applies verbatim to Rule 90.
52	Homotopy type theory / loop-space framing	1	Kil	All types are hSets; the statement 0-truncates to the plain first-order sentence.
53	Latent-space geometry to conserved window functions	1,2	Kil	Obstruction C (single-column blindness) at step 0, width-independently, over $\mathbb{Z}/2, \mathbb{Z}/3$ to $w = 12$; ML half was the candidate generator for a class killed wholesale, so no generator can rescue it.
54	Ladder tightening: right-wedge constraint	1	Kil	Constraint already imposed in the code; every verdict NONEMPTY with or without it.
55	Ladder tightening: boundary pin	1	Kil	Genuine constraint, prunes 39–58%, but rung 1 proved $\text{plain}(R+1) \subseteq \text{pin}(R) \subseteq \text{plain}(R)$: buys less than one column.
56	Ladder tightening: bounded-universal Q	1	Kil	NONEMPTY to $Q = 14$; constraint and available freedom both grow as 2^Q .
57	Hyperbolic lightcone geodesic embedding	1	Kil	Column-blind: overwriting column 0 moves the statistic by 10^{-4} – 10^{-7} against a 96% positive control.
58	Curvature of a multilinear relaxation	1,2	Kil	Column-blind to machine epsilon; also a category error (no Riemannian metric on a max-plus semiring).
59	Spectral dimension of a Dirac operator on the shift algebra	1	Kil	Column-blind, $O(1/W)$ to three digits; Rule 90 induces the same algebra with the opposite P1 answer.
60	OTOC / spectral form factor	1	Kil	Premise false: Rule 30 is non-invertible, so no unitary evolution operator or RMT ensemble exists.
61	2-adic/Mahler expansion, unique ergodicity under Haar measure	2	Kil	Hypothesis false in one line: δ_0 and uniform Bernoulli are both invariant.
62	Braid-group word length of defect world lines	1	Kil	No braid word exists: Rule 30 defects merge and annihilate (forced by non-invertibility).
63	Algebraic immunity of f_t as an AC0 lower bound	3	Kil	AC0 claim false: AND_n has maximal degree, algebraic immunity 1, and a depth-2 AC0 circuit.
64	AC0/Smolensky size bound from the ANF degree of f_t	3	Kil	Refuted by the same AND_n counterexample; also the trivial regime ($n = 2t + 1$ variables).
65	Straight-line-program length along the single-seed line	3	Kil	Category error: on one fixed point every measure of this kind is identically zero.
66	Sensitivity $s(f_t) \geq t$ along the single-seed path	3	Kil	Numerically false: measured sensitivity sits below t and below the random-input maximum throughout.
67	Ground-truth centre-column generator and OEIS gate	infra	Inf	Bit-parallel generator matched against OEIS A051023; gates every arm’s pipeline.
68	Modal compute charter and admission guard	infra	Inf	\$200 ceiling, fail-closed guard; never entered (every result records \$0).
69	Exclusion-guided search runbook	infra	Inf	An LLM-driven candidate search in which every rejected candidate becomes a typed exclusion later candidates must differ from, keeping the space auditable rather than repeating known-dead structures.
70	Overnight autonomous run: prompt and fences	infra	Inf	Fenced session producing rows 42–47; compliance verified by mtime, one disclosed exception.
71	Settled-knowledge digest for downstream sessions	infra	Inf	Read-only survey of what is proved, dead, and off-limits.
72	Zero-tail note: manuscript, figures, Lean/SMT certificates	infra	Inf	Packaging for row 25 plus a hostile proof audit; novelty over Jen 1986 confirmed componentwise.
73 [†]	Preimage-tree / backward-dynamics descent	1	Kil	Column-blind at step 0; premise was a boundary confusion; object is the pin restated.
74 [†]	Ore ladder uniformity by induction on rungs	1	Kil	Explicit family has strict Ore order at every level; only the induction sub-route dies (row 46 stays open).

#	Route	P	St.	Kill reason
75 [†]	Stroboscopic descent for $p = 2$	1	Opn	Support grows exactly 2 per side under F^2 ; no width-shaped quantity can decrease.
76	S-adic decomposition + Morse-Hedlund complexity	1	Kil	Killed by obstruction H; bound corrected upward $\sim 3,100\times$ on remerge (linear in prefix length, not $n_{\max}$).
76 [†]	Row 41 d -uniform survivor invariant (second stream)	1	Opn	Time-reindexed recurrence found, but natural truncation is vacuous by a light-cone argument.
77 [†]	Ladder realizability / half-plane slip	1	Prv	Slip extends to a full half-plane; row 7’s ladder mode (i) can never return EMPTY at any depth, retiring it entirely.
78 [†]	P2 nontrivial density bounds	2	Kil	Gap bound is linear in t ; already implied by row 47’s Theorem A.
79 [†]	Orbit-closure patch census	2	Ref	Checkerboard is a Rule 30 temporal fixed point; R8’s target fails if it lies in Y (measured to $K = 6$).
80 [†]	Rowland-Yassawi nonlinear analog	2	Neg	Only arrow of the R-Y chain that avoids linearity delivers just a simplex.
81 [†]	Resolution lower bound for c_n	3	Kil	Metric mis-specified (obstruction I).
82 [†]	Trace subshift soficness	1,2	Neg	Width-1 object is the full shift by construction (obstruction B); width-2 reduction gives only a lower bound.
83 [†]	Conserved non-additive window functions	1,2	Kil	Column-blind width-independently; exhaustive absence to $w = 12$; Rule 90 gives identical answers.
84 [†]	Jen 1986 provenance	infra	Prv	Full text obtained and read (JSP 43:219–242); clause (ii) is Thm. 7a (p. 236), Rule 30 fails every condition; Thm. 2b’s citation above independently confirmed.
85 [†]	Sub-quadratic derivation family for c_n	3	Kil	R9’s inverted kill did not fire; best family is exact $\Theta(n^2)$, $2.95\text{--}3.88\times$ under baseline (obstruction I).
86 [†]	Nonabelian conserved transfer invariant	1,2	Kil	1,988 apparent invariants passed an initially too-loose filter; corrected, then closed generally: any monoid-valued transfer is $\leq \log_2 M $ bits about the orbit.
87 [†]	Polynomial Calculus degree for c_n	3	Kil	Degree exactly 2 (Rule 30) vs. 1 (Rule 90), both constant in n ; fails the Rule 90 filter honestly (obstruction I).
88 [†]	2-adic Newton-polygon / valuation argument	1	Kil	No closed functional equation exists (the pin’s nonlinear term is a Hadamard product, not a ring operation).
89 [†]	Cryptanalytic bias detection on the lone-seed trace	—	NBF	BM complexity, correlation attack, NIST STS, autocorrelation to 2^{21} ; smallest corrected $p = 0.627$.
90 [†]	ML-guided conjecture mining	1	Ref	Trained to depth 20000: all 30 surviving candidates break within 38 steps of the training boundary; only the known OR-latch pin survives.
91	OpenEvolve LLM-driven search for sub-quadratic $c(n)$	3	NR	Sanity gate failed on re-verification: wall-clock instrument credited the baseline with beating itself. Instrument rebuilt and validated as row 92; the search itself has still not been run with it.
92	Deterministic fuel-counter evaluator (fixes row 91’s instrument)	3	Prv	Row 12’s WASM fuel meter was read and rejected for reuse (WASM ops are genuinely $O(1)$; Python bignum ops are not). Word-RAM AST counter recovers exponent 1.99946 at $r^2 = 0.9999999$; publication claim withdrawn as prior art: counter-beats-clock fitness [1], proportional-width bignum charging [2], and deterministic counters replacing wall-clock timing in general [7] are each already owned.
93	Bounded exact circuit synthesis for $n \rightarrow c(n)$	3	Prb	$k_{\min}(4) = 5$ unconditional; $k_{\min}(5) \geq 7$ conditional; strong outcome did not fire; P3 untouched.

7 Part V: Scope, the unifying fact, and open problems

Problems 1, 2 and 3 are open and nothing here changes that. Part I settles eventual period one completely, for a class strictly larger than the lone seed, and is the paper’s one unconditional advance on the center-column question. Part III’s limitation theorems close a specific mechanized route to period two. Part II’s census and lead delimit period two without deciding it. The unifying fact is the Rule 90 filter: the OR nonlinearity is the entire difference, every proven result routes through it, and every failed route does not.

Remark 6.2 sharpens that statement. The OR has two branches and they are not two aspects of one behaviour: in the right light-cone frame it feeds a running parity and the diagonal periods double every 2.4 steps of depth, while in the left frame it is self-referential and absorbing and the period doubles only at depths 3, 8, 29, 400, 87 867, $\dots$, giving an ordered region Rule 90 does not have. So the unifying fact is better stated as: the OR makes Rule 30 *two-phased*, and the center column is the one place the prize question asks about that lies in the disordered phase. We record this as a reorientation, not a result. No proof of the glide identity is offered; the ordered region’s period is unbounded, only very slowly so; and we know of no argument that carries a left-region theorem rightward across the interface — and no measured centre-only propagation has bridged the gap from the other side.²

The concrete open leads are three, one fewer than when this paper was first drafted: the comparison asking whether the extremal forcing states of Observation 4.3 form a bounded-period family across n has been run and is negative, as recorded after Conjecture 4.4, so the fixed-family cocycle argument it would have enabled is not in range. What remains is a Rule 90 carry-kernel census as the missing soundness control on the period-two thread; the separator SEP itself, the one open arrow of (6); and a second, independent route to SEP via mortality of an exact reversed-diagonal queue construction. On that route, every fixed-radius end window is proved nonclosed as a transition congruence, however it is augmented by queue length, endpoint/event data, or the exact total dihedral action of the interior symbols: two interior blocks collapsing to the same group element are not dynamically interchangeable, and no bounded summary of that kind can tell them apart. The one candidate escape that specifically avoids this obstruction, a finer congruence retaining the full intermediate scan trace rather than its group-collapsed summary, turns out to close the family rather than escape it: beyond an exhaustive census of 3,015,168 words showing no class merges, it is injective at every length, since each symbol acts on the scan states as a permutation and distinct normalized symbols act differently somewhere. The induced right congruence is therefore the identity and the block monoid is free, so no bounded lossless summary of an interior block exists at any radius, of any shape. Two further candidate escapes (a paired/relative invariant over synchronized copies; a monotone potential not built from any finite quotient) were tested and fail.

The same route’s remaining lemma has since been reduced twice and then closed as a mechanism. The lemma needed is the pair of bounds $s_2(W) \leq n$, $s_3(W) \leq n + 1$ on the forced continuation length of a hard-core source of length n . A deterministic greedy matching certifies it through length 22; the greedy construction can be weakened to a per-row sensitivity claim, and that in turn holds after quotienting the newest-cut group by its centre, which for this D_8 coincides with the commutator subgroup. The

²A run of ones is sufficient but *not* necessary for centre-only forcing: $c(t) = \dots = c(t + k - 1) = 1$ forces $\text{col}_{-k}(t) = c(t + k) \oplus (k \bmod 2)$, but of the 256 length-8 centre words, 116 force a cell deeper than their run-of-ones prefix. The shortest witness is 01010, which forces $\text{col}_{-4} = 0$ while leaving depths 1, 2, 3 free; the mechanism is the relation $bd = 0$ between two *unknown* right-column cells $b = \text{col}_1(t)$, $d = \text{col}_1(t + 2)$, which a known/unknown propagation cannot see. Rule 90 leaves col_{-4} free on the same word, so this is not generic left permutivity. Exact reach at a fixed depth k is decided by enumerating the 2^k initial right rows; on the lone seed the maxima are 15, 27, 27, 31, 31 over horizons $t < 100, 500, 1000, 2000, 4096$ with depths tested through 63, attained at times whose run length is 0 or 1. Two earlier drafts are retracted: a “one-column reach” that saturated the wrong argument of the OR, and an “exactly when” whose $O(\log t)$ consequence does not follow. We claim no asymptotic reach bound here; the finite tables are evidence, not a growth law.

resulting two-bit invariant is exactly the (α, β) pair of the existing affine coordinates, with the quadratic coordinate γ the kernel and never required; neither bit alone suffices, so two is the exact size. That strengthened claim holds on 72,351 legal rows through source length 21, clearing the length at which an earlier pointwise-derivative certificate broke, and is tight on exactly two words. Nevertheless the reachability form of the claim does not localize. The set of suffix configurations whose invariant freezes along the nested chain is nonempty, sparse, and never legal, but the freeze is independent of the prefix — every scenario in the chain zeroes it — while the continuation length depends on the prefix and its first forced symbol is almost entirely a function of it. Ten candidate local mechanisms relating the two were refuted by explicit witnesses, and on the restricted class the bound required is strictly stronger than the target itself. This is a further instance of obstruction D, the missing composition law: there is no composition law for the continuation length, so concatenation cannot be made to lower it. This route is reported for completeness and is not otherwise used above.

8 Reproducibility

All computational claims are reproduced by scripts in the accompanying archive, seeded and cross-checked; the exhaustive audits and SMT certificates supplement but do not replace the unbounded proofs of Part I. The census (`fastdk_core.py`) and ladder (`run_rung0.py`, `rung1.py`) ship with their own cross-checks against brute force or an independent implementation; the needle analysis is cross-checked the same way and detailed in its result document. The measurements of Remark 6.2 are produced by `ordered_wedge.py`, which computes the same object twice — once by the plain update in (t, x) and once by each moving-frame recurrence — and whose test module asserts that the two agree cell by cell, that no glide length outside the multiples of 16 has a wedge, and that the Rule 90 control fails the bounded-period claim. The full corpus — scripts, logs, and the result documents underlying the register — is deposited separately (Zenodo DOI: [to mint]), and the exhaustive per-route register is the supplement.

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